4
COMBINATIONAL LOGIC

4.1 Basic Concepts

Digital logic circuits can be classified into two types: combinational and sequential. A combinational circuit is designed using logic gates in which application of inputs generates the outputs at any time. An example of a combinational circuit is an adder, which produces the result of addition as output upon application of the two numbers to be added as inputs.

A sequential circuit, on the other hand, is designed using logic gates and memory elements known as *flip-flops.* Note that the flip-flop is a one-bit memory. A sequential circuit generates the circuit outputs based on the present inputs and the outputs (states) of the memory elements. The sequential circuit is basically a combinational circuit with memory. Note that a combinational circuit does not require any memory (flip-flops),whereas sequential circuits require flip-flops to remember the present states. A counter is a typical example of a sequential circuit. To illustrate the sequential circuit, suppose that it is desired to count in the sequence 0, 1, 2,3, 0, 1,… and repeat. In binary, the sequence is 00, 01, 10, 11, 00, 01, …, and so on. This means that a two-bit memory using two flip-flops is required for storing the two bits of the counter because each flip-flop stores one bit. Let us call these flip-flops with outputs A and B. Note that initially $A = 0$ and $B = 0$. The flip-flop changes outputs upon application of a clock pulse. With appropriate inputs to the flip-flops and then applying the clock pulse, the flip-flops change the states (outputs) to $A = 0$, $B = 1$. Thus, the count to 1 can be obtained. The flip-flops store (remember) this count. Upon application of appropriate inputs along with the clock, the flip-flops will change the status to $A = 1$, $B = 0$; thus, the count to 2 is obtained. The flip-flops remember (store) the present value of the count at the outputs until a common clock pulse is applied to the flip-flops. The inputs to the flip-flops are manipulated by a combinational circuit based on A and B as inputs. For example, consider $A = 1$, $B = 0$. The inputs to the flip-flops are determined in such a way that the flip-flops change the states at the clock pulse to $A = 1$, $B = 1$; thus, the count to 3 is obtained. The process is repeated.

In this chapter, analysis and design of typical combinational circuits will be covered.

4.2 Analysis of a Combinational Logic Circuit

Analyzing a combinational circuit means that the circuit is given, the output equation (s) in terms of the input (s) and the truth table need to be determined. Hence, the combinational logic circuit can be analyzed by (i) first, identifying the number of inputs and outputs, (ii) expressing the output functions in terms of the inputs, and (iii) determining the truth table for the logic diagram.

Fundamentals of Digital Logic and Microcontrollers, Sixth Edition. M. Rafiquzzaman.

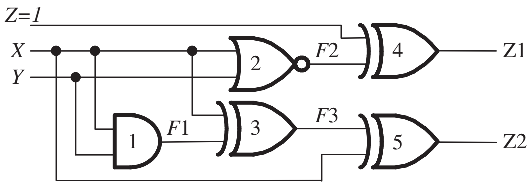


FIGURE 4.1 Analysis of a combinational logic circuit.

As an example, consider the combinational circuit in Figure 4.1 There are three inputs (X, Y, and Z) and two outputs ($Z1$ and $Z2$) in the circuit. Let us now express the outputs $F1$ and $F2$ in terms of the inputs. The output $F1$ of the AND gate #1 is $F1 = XY$. The output $F2$ of NOR gate #2 can be expressed as $F2 = \overline{X+Y}$. The output of the XOR gate #3 is

$$F3 = X \oplus F1 = (X \oplus XY)$$

Because one of the inputs of the XOR gate #4 is 1, its output is inverted. Therefore,

$$Z1 = \overline{F2} = X + Y$$

Finally,

$$Z2 = X \oplus F3 = X \oplus (X \oplus XY)$$

Therefore,

$$\begin{aligned} Z2 &= X \oplus (X(1 \oplus Y)) \\ &= X \oplus X\overline{Y} \text{ since } 1 \oplus Y = \overline{Y} \\ &= X(1 \oplus \overline{Y}) \\ &= XY \text{ since } 1 \oplus \overline{Y} = Y \end{aligned}$$

Another way of determining $Z2$ is provided below:

$$Z2 = X \oplus F3 = X \oplus (X \oplus XY) = X \oplus X \oplus XY = 0 \oplus (XY) = XY$$

The truth table shown in Table 4.1 can be obtained by using the logic equations for $Z1$ and $Z2$.

4.3 Design of a Combinational Circuit

Design of a combinational circuit is basically a reverse process of analyzing the circuit. Note that the term *design* means that a designer needs to come up with a logic circuit following certain criteria such as minimum cost and highest speed. In this book, logic circuits are designed based on minimum cost. That is why minimization techniques are used to reduce the number of logic gates in a circuit. This will provide lower implementation cost. In summary, a combinational circuit can be designed using three steps as follows:

1) **Truth Table:** If the truth table is given, go to step 2; else, determine the inputs and outputs from problem definition and then derive the truth table.

TABLE 4.1 Truth table for Figure 4.1 with input, $Z = 1$

Inputs		Outputs	
X	Y	$Z1$	$Z2$
0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	1

TABLE 4.2 Truth Table for ***F***

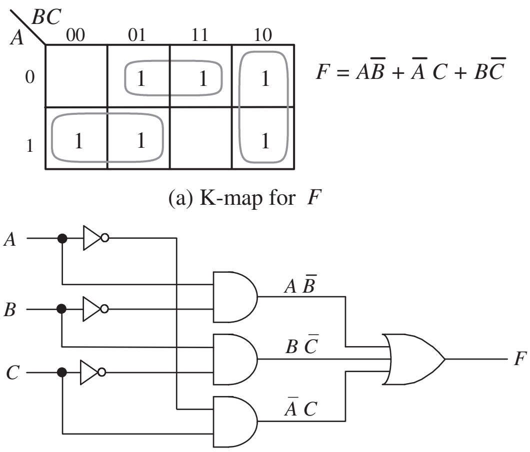
A	*B*	*C*	*F*
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

2) **Minimization:** Minimize the number of inputs (literals) in order to express the outputs. This reduces the number of gates and thus the implementation cost. In order to accomplish this, first use K-map(s). Note that the number of K-maps is determined by the number of outputs. If the output(s) cannot be simplified using K-map(s), then obtain output equation(s) from the truth table directly, and then simplify using Boolean identities. Simplification is also sometimes possible using the K-map(s) first and then using Boolean identities.
3) **Logic diagram or schematic:** Draw the logic diagram or schematic using logic gates. Note that since the terms *Logic diagram* and *Schematic* mean the same thing, they will be used interchangeably throughout this book.

In order to illustrate the design procedure, consider the following example. Suppose it is desired to design a combinational circuit with three inputs (A, B, and C) and one output F. The output F is one if A, B, and C are not equal ($A \neq B \neq C$); $F = 0$ otherwise. The three steps for designing the circuit is provided next:

Step 1 Truth Table: Since the truth table is not given, the number of inputs and outputs are determined. There are three inputs (A, B, and C) and one output, F. The truth table is then obtained from problem definition and is shown in Table 4.2.

Step 2 Minimization: Output, F in the truth table of Table 4.2 is simplified using a K-map (Figure 4.2(a)). The output equation is obtained as $F = A\overline{B} + \overline{A}C + B\overline{C}$



(a) K-map for F

(b) Logic Diagram for the output, F

FIGURE 4.2 K-map and the logic diagram for ***F***.